Test report 21 – hull/minkowski, STL import, collision asserts, user feedback, rename to mcad

Plan 21 (phases A–C), user feedback (phase D), rename (phase E): all commits since `852fbc1` · 2026-10-25 · Branch ccr-4837f657-244mcz · Commit 0ea0d78 · mcad 0.3.0 · rustc 1.97.0 (2d8144b78 2026-07-07) · 4 cores. Measurements with the release binary without daemon (`tools/report/report21-data.py`), test runs on the same commit.

hull/minkowski vs. formula
20 / 20
within 10^{-5} relative; 3 cases without a special case → warning

STL: B-rep up to 500 faces
1.3 s
build + Boolean at 445 faces; 4,646 faces: 86 s

Assert cases
23 / 23
kind and value as expected; export on violation exits with 4

cargo test --workspace
658 / 658
clippy clean, fmt clean

Vitest · Playwright
246 · 149 / 149
browser with Wasm kernel

Tauri E2E
20 / 20
desktop app with native kernel

OpenSCAD · STEP · everyday
36/36 · 64/64 · ok
comparison, OCCT check, 19 models

1 Scope

Phase	Commit	Title
A	`83ee5b3`	Kernel: outward offset (Minkowski sum with a sphere), exact or error
A	`71e5bce`	Kernel geometry: exact 2D hull from lines and arcs, convex hull in 3D
A	`b97c365`	hull() and minkowski(): exact special cases in 2D and 3D, otherwise a warning with the reason
A	`a70dc9a`	minkowski() with one child is the child, empty hull()/minkowski() without warning
B	`b2b9768`	Kernel: triangle meshes without B-rep, STL parser and mesh preparation (cad-stl)
B	`8847eb2`	Convex hull in 3D: closed also for nearly coplanar points
B	`b6e8906`	import("x.stl"): STL as B-rep or as plain mesh (Plan 21, phase B)
B	`f017264`	UI: STL as data file (upload, examples, references in the text)
Editor	`2d3451e`	Editor: no more inline sliders for number literals
C	`c585002`	Collision asserts: assert_keepout/disjoint/intersect/touches, cad export aborts (Plan 21, phase C)
C	`0e20aad`	UI: keep-out volume hatched and hideable, assert results in the view, export confirmation
C	`bf44805`	Tauri E2E T19: keep-out volume, framing a violation, export confirmation in the desktop app
C	`235d4af`	Asserts: contact exactness per contact kind, CLI shows intended intersections in green
D	`a86a21b`	JSON schema 2: selected_edges as polylines everywhere, uniform fields in outline
D	`cbf2963`	-D without module context: set top-level assignments (like openscad -D)
D	`1d530de`	-m 'module(arguments)': module context with arguments as in the code
D	`91f8ef6`	slot(p1, p2, r): slot as exact 2D primitive
D	`8ac92d6`	STEP export: one named body per part and object, --merge for a single one
D	`3164682`	Kernel errors with location: tangentially touching operands with line and position
E	`4db585b`	Rename cad → mcad: binary, program name, file extension .mcad (version 0.3.0)
E	`004d30f`	Tauri E2E: binary mcad, T20 window title, app ID and .scad project
Report	`0ea0d78`	verify.py: removed extra parenthesis left over from the rename

User decisions for Plan 21: `hull/minkowski` only in exact special cases, otherwise as before a warning with unchanged children (no approximation); STL import with an automatic B-rep/mesh limit and override `as = "solid" | "mesh"`; asserts: violation = warning, `mcad export` aborts (unless `--ignore-asserts`), `assert_keepout` clearly visible in the UI.

2 Phase A: Offset, hull, minkowski

Kernel offset (spike)

`Kernel::offset(s, d, tag)` is the Minkowski sum of a body with a sphere: facade `sc_offset` via `SmApiOffsetBrepFull` (outward, rounding, self-intersection handling). The spike (state `83ee5b3`, volume from the mass properties against the Steiner formula) shows where SMLib is exact: convex edges become cylinders and corners become spheres, as long as the faces meet at right angles at the corners. On the tetrahedron the surfaces are correct (4 planes, 6 cylinders, 4 spheres with exact centers and radii), but the trim curves lie up to 0.33 mm off the edges (volume $+4.5{\cdot}10^{-4}$ vs. Steiner); on the 45° wedge SMLib fails. The facade therefore checks the distance of the result boundary to the body (9 points per edge, face center) and the trim curves against the 3D edges (10^{-6} of the size), and otherwise reports "offset is not exact".

Body (r = 2)	Rounding without self-intersection (mode 0)	Rounding with self-intersection (mode 2, chosen)	Substitute: face offset + fillet	Faces (mode 2)
Cuboid 10×20×30	error (self-overlapping)	ok, $+3.0{\cdot}10^{-8}$, 184 ms	error (not exact: boundary up to 1.46 mm off the offset distance)	12 cylinder, 6 plane, 8 sphere
Cuboid rotated	ok, $+1.6{\cdot}10^{-8}$, 108 ms	ok, $+2.7{\cdot}10^{-8}$, 180 ms	error (not exact: boundary up to 1.46 mm off the offset distance)	12 cylinder, 6 plane, 8 sphere
Cylinder r 5, h 10	ok, $+1.7{\cdot}10^{-7}$, 36 ms	ok, $+1.7{\cdot}10^{-7}$, 59 ms	error (not exact: boundary up to 0.83 mm off the offset distance)	1 cylinder, 2 plane, 2 revolution
Sphere r 5	ok, $-2.8{\cdot}10^{-7}$, 15 ms	ok, $-2.8{\cdot}10^{-7}$, 20 ms	error (fillet fails)	1 sphere
L profile (concave edge)	error (not closed)	ok, $-5.3{\cdot}10^{-8}$, 372 ms	error (not exact: boundary up to 1.46 mm off the offset distance)	17 cylinder, 8 plane, 10 sphere
filleted cuboid (r 1)	ok, 911 ms	ok, 881 ms	error (fillet fails)	12 cylinder, 6 plane, 8 sphere
Tetrahedron	error (not exact: trim curves up to 0.33 mm off the edges)	error (not exact: trim curves up to 0.33 mm off the edges)	error (not exact: boundary up to 5.98 mm off the offset distance)	–
45° wedge (extrusion)	error (self-overlapping)	error (self-overlapping)	error (not exact: boundary up to 3.60 mm off the offset distance)	–
45° wedge (polyhedron)	error (self-overlapping)	error (self-overlapping)	error (not exact: boundary up to 3.60 mm off the offset distance)	–

Column "substitute": face offset (faces extended) followed by filleting all edges with r – not viable on any body except the cylinder and the rotated cuboid, therefore discarded. Raw data `reports/tmp/report21/phase-a/offset-spike.txt`.

Language and example

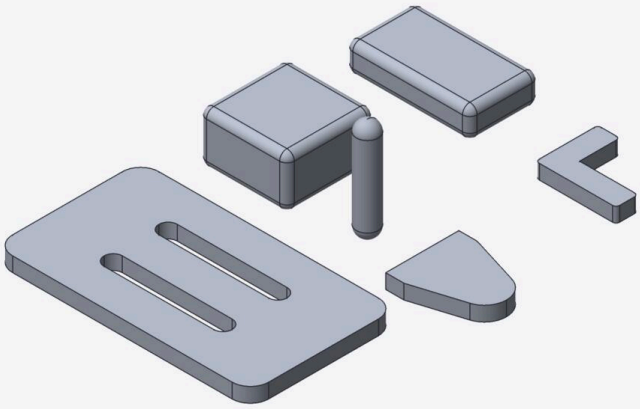
`hull()` and `minkowski()` choose the case by their children (transformations, groups, unions and nested hulls are looked through; spheres/cylinders/arcs only under similarity transformations). Exact special cases:

Operation	Special case	Result
`hull` 2D	every region made of lines and circular arcs (text, SVG: B-splines via biarcs)	lines + arcs via the support function, transitions at the common tangents in closed form
`minkowski` 2D	profile + circle; two convex profiles (one of them a polygon)	offset or convex sum
`hull` 3D	planar bodies; spheres/cylinders/cones on one axis; parallel prisms of equal height; equal spheres	polyhedron; solid of revolution of the 2D hull of the half sections; extruded 2D hull; hull of the centers + offset r
`minkowski` 3D	body + sphere; `linear_extrude` + `cylinder` (parallel prisms); two convex planar bodies; one child	offset; prism of the 2D sum; polyhedron; the child
otherwise	e.g. spheres of different radii not on one axis, tetrahedron + sphere, torus, `$fillet` primitives	warning "hull: <reason>; children are shown unchanged" (German output: „hull: <Grund>; Kinder werden unverändert gezeigt“)

```
module langloch(l = 20, d = 6) {
    hull() {
        circle(d = d);
        translate([l, 0]) circle(d = d);
    }
}

module platte(w = 60, t = 30, h = 4, r = 5) {
    difference() {
        linear_extrude(h) hull() {
            for (x = [0, w], y = [0, t])
translate([x, y]) circle(r);
        }
        for (y = [t / 4, 3 * t / 4])
translate([w / 4, y, -1]) linear_extrude(h + 2)
langloch(w / 2, 5);
    }
}

platte();
translate([0, 50, 0]) minkowski() {
    cube([20, 20, 10]);
    sphere(2);
}
translate([40, 52, 0]) hull() {
    translate([0, 0, 3]) sphere(3);
    translate([0, 0, 25]) sphere(3);
}
translate([60, 52, 0]) hull() {
    translate([-5, -10, 0]) cube([5, 20, 4]);
    translate([12, 0, 0]) cylinder(r = 6, h =
4);
}
translate([0, 90, 2]) hull() {
    for (x = [0, 30], y = [0, 15], z = [0, 4])
translate([x, y, z]) sphere(2);
}
translate([50, 90, 0]) minkowski() {
    linear_extrude(3) polygon([[0, 0], [20, 0],
[20, 4], [4, 4], [4, 15], [0, 15]]);
    cylinder(r = 1, h = 1);
}
```



`examples/spec/hull_minkowski.mcad`, `mcad render` (iso): plate as 2D hull of four circles with slots, rounded block (cube ⊕ sphere), handle as capsule, tab (cuboid + cylinder of equal height), cushion of eight equal spheres, strip from `linear_extrude` ⊕ `cylinder`

```
$ mcad check examples/spec/hull_minkowski.mcad
examples/spec/hull_minkowski.mcad: 0 Fehler, 0
Warnungen
(Exit 0)
```

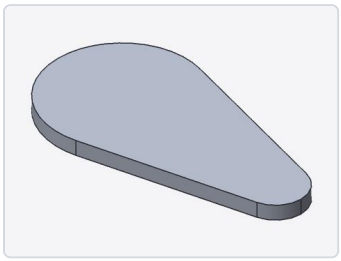
Measurements vs. formula

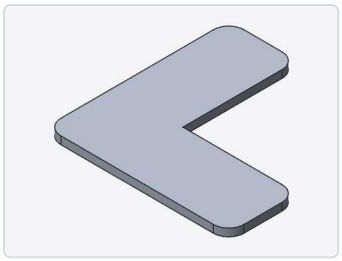
Volume via `mcad measure --volume` (exact B-rep, mass properties of the kernel); 2D cases extruded by 1, so the volume is the area. Time = full CLI call without daemon (parsing, evaluation, kernel, volume), median of 3 runs.

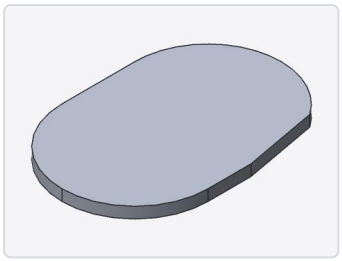
Case	Operation	Formula	expected	measured	rel. dev.	call	Special case or warning
Stadium r 3, L 10	`hull`	$\pi r^2 + 2rL$	88.2743	88.2743	$+9.6\cdot10^{-9}$	66 ms	
Circles r 5 and r 2, distance 12	`hull`	belt area	132.1920	132.1920	$+7.2\cdot10^{-9}$	71 ms	
Square 10 + circle r 5 on the edge	`hull`	$100 + 12.5\pi$	139.2699	139.2699	$+6.7\cdot10^{-9}$	63 ms	
4 circles r 2 on rectangle 20×10	`hull`	$24\cdot14 - (4 - \pi)\cdot4$	332.5664	332.5664	$+3.9\cdot10^{-9}$	64 ms	
Hull of the glyph "G" (size 20)	`hull`	–	–	226.0389	–	67 ms	
Rectangle 10×20 ⊕ circle r 2	`minkowski`	$A + Pr + \pi r^2$	332.5664	332.5664	$+3.9\cdot10^{-9}$	63 ms	
L profile ⊕ circle r 2 (concave corner)	`minkowski`	$A + Pr + (5\pi/4 - 1)r^2$	346.7080	346.7080	$+4.8\cdot10^{-9}$	70 ms	
Square 10 ⊕ triangle 4×3	`minkowski`	176	176.0000	176.0000	$-1.6\cdot10^{-16}$	46 ms	
Capsule r 3, L 10	`hull`	$\pi r^2 L + 4/3\pi r^3$	395.8407	395.8410	$+7.9\cdot10^{-7}$	123 ms	solid of revolution
Capsule rotated [30, 40, 0]	`hull`	$\pi r^2 L + 4/3\pi r^3$	395.8407	395.8411	$+1.2\cdot10^{-6}$	131 ms	solid of revolution
Spheres r 5 and r 2, distance 10	`hull`	cap + truncated cone + cap	723.7182	723.7181	$-1.5\cdot10^{-7}$	146 ms	solid of revolution
Cylinder r 3 h 10 + sphere r 3 on top	`hull`	$\pi r^2 h + 2/3\pi r^3$	339.2920	339.2921	$+4.1\cdot10^{-7}$	100 ms	solid of revolution
two cubes 10 at distance 20	`hull`	$30\cdot10\cdot10$	3,000.0000	3,000.0000	$-3.0\cdot10^{-16}$	53 ms	polyhedron (6 faces)
Cube 10 + "point" 5 above the top face	`hull`	$1000 + 100\cdot5/3$	1,166.6667	1,166.6700	$+2.9\cdot10^{-6}$	57 ms	polyhedron (10 faces)
3 cylinders r 3 h 2	`hull`	$h\cdot(\text{triangle} + Pr + \pi r^2)$	716.5487	716.5487	$+6.5\cdot10^{-9}$	75 ms	prism
Cuboid 10×10×5 + cylinder r 5 h 5	`hull`	$h\cdot(200 + 12.5\pi)$	1,196.3495	1,196.3495	$+4.8\cdot10^{-9}$	83 ms	prism
8 spheres r 2 at the corners of 20×10×5	`hull`	cuboid ⊕ sphere (Steiner)	2,873.3333	2,873.3334	$+2.4\cdot10^{-8}$	442 ms	hull of the centers + offset 2
Cuboid 10×20×30 ⊕ sphere r 2	`minkowski`	Steiner	11,187.4926	11,187.4929	$+3.0\cdot10^{-8}$	278 ms	offset 2
rotated cuboid ⊕ sphere r 2	`minkowski`	Steiner	11,187.4926	11,187.4929	$+2.6\cdot10^{-8}$	304 ms	offset 2
Prism 10×20×5 ⊕ cylinder r 2 h 1	`minkowski`	$6\cdot(A + Pr + \pi r^2)$	1,995.3982	1,995.3982	$+3.9\cdot10^{-9}$	79 ms	prism
Cube 10 ⊕ cuboid 1×2×3	`minkowski`	$11\cdot12\cdot13$	1,716.0000	1,716.0000	$-2.7\cdot10^{-16}$	52 ms	prism
Tetrahedron ⊕ sphere r 1	`minkowski`	–	–	170.3319	–	331 ms	**Warning:** Offset 1: offset is not exact: trim curves up to 0.166954 mm off their edges (SMLib approximates the rounding at corners whose faces do not meet at right angles); children are shown unchanged
3 spheres r 3/2/1 not on one axis	`hull`	–	–	150.7968	–	262 ms	**Warning:** spheres with different radii not on one axis; children are shown unchanged

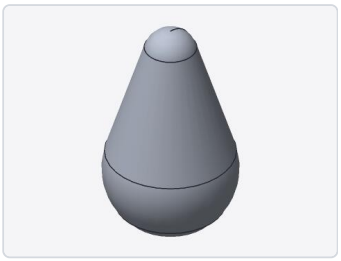
Case	Operation	Formula	expected	measured	rel. dev.	call	Special case or warning
4 equal spheres r 2 in one plane	`hull`	–	–	134.0418	–	327 ms	**Warning:** equal spheres with centers in one plane; children are shown unchanged

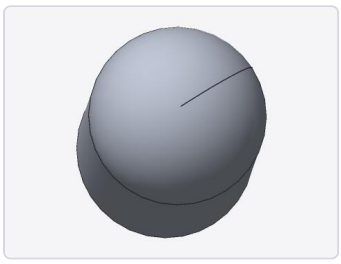
Deviations: 2D hulls and prisms ≤ 10^{-8} (integration); solids of revolution (capsule, spheres, dome) 10^{-7}–10^{-6} – accuracy of the volume integration of surfaces of revolution, the geometry itself is exact (STEP: `SURFACE_OF_REVOLUTION`, no B-splines); cube + "point": the point is a cube with 10^{-4} edge length (+2.9·10^{-6} is its contribution).

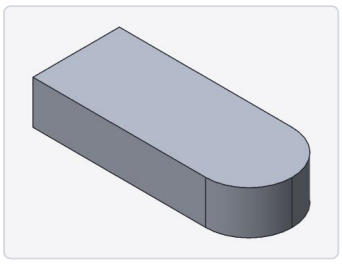

Circles r 5 and r 2, distance 12

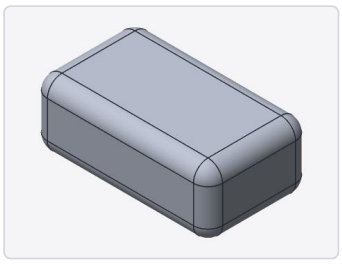

L profile ⊕ circle r 2 (concave corner)

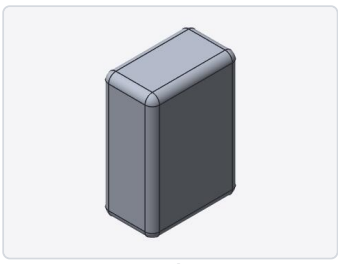

Hull of the glyph "G" (size 20)

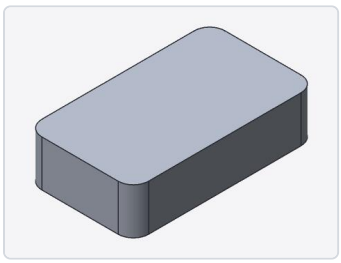

Spheres r 5 and r 2, distance 10

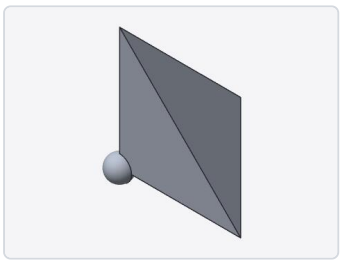

Capsule rotated [30, 40, 0]

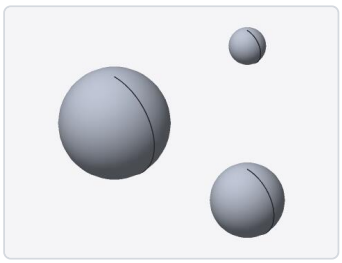

Cuboid 10×10×5 + cylinder r 5 h 5

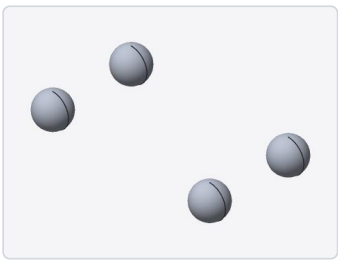

8 spheres r 2 at the corners of 20×10×5


Cuboid 10×20×30 ⊕ sphere r 2


Prism 10×20×5 ⊕ cylinder r 2 h 1


Tetrahedron ⊕ sphere r 1 – warning, children unchanged


3 spheres r 3/2/1 not on one axis – warning, children unchanged


4 equal spheres r 2 in one plane – warning, children unchanged

Limitations

- Offset (and thus `minkowski` with a sphere) only on bodies whose faces meet at right angles at the corners; tetrahedra, wedges, oblique polyhedra → warning, children unchanged.
- Equal spheres with centers in one plane (rounded plate) are not a special case (the hull of the centers is flat, an offset of it does not work) – case "4 equal spheres in one plane" above; workaround: `linear_extrude` of a 2D hull with `$fillet`.
- `rotate_extrude` children, torus and `$fillet` primitives in `hull`: warning.

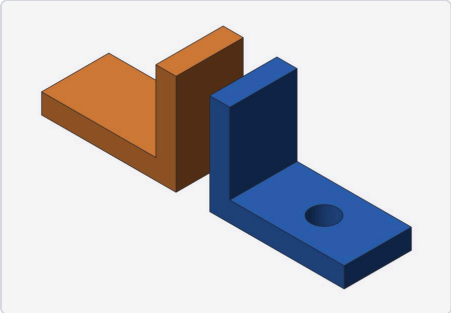
3 Phase B: `import("x.stl")`

Language and modes

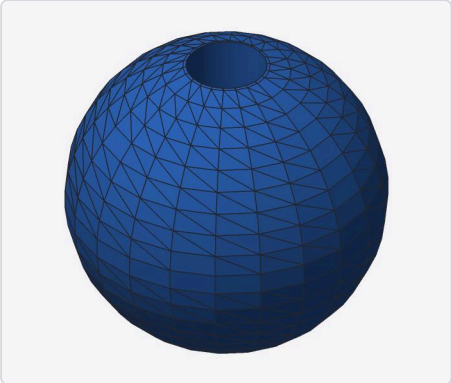
```
import("../stl/winkel.stl");                    // closed, ≤ 500 faces: B-rep (Booleans work)
import("platine.stl", as = "mesh");             // plain mesh (display, export, measuring)
import("teil.stl", center = true, scale = 25.4); // inch → mm, bounding box around the origin
```

Mode	automatic if	works	does not work
B-rep (`solid`)	closed, manifold, ≤ 500 faces after merging	everything: Booleans, `fillet`/`$fillet`, exact measuring, STEP	–
Mesh (`mesh`)	> 500 faces, open, non-manifold	display, transformations, `color`, parts, `hull` (≤ 2000 vertices), STL/3MF, STEP faceted, bounding box/volume/area/section, `--clearance`, `--interference` (yes/no)	Booleans, `fillet`/`chamfer`/`shell`, `minkowski`, `thicken` – error or warning with the hint `as = "solid"`

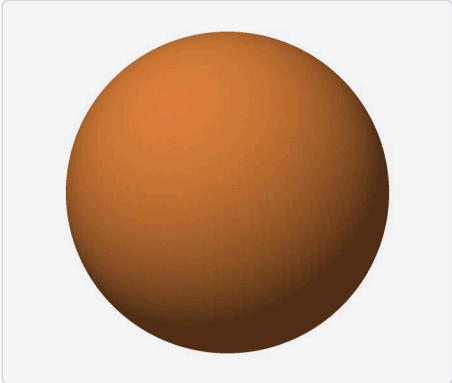
Preparation in every mode (`cad-stl`): weld vertices (10^{-6} of the diagonal), discard NaN, degenerate and duplicate triangles, orient consistently outward, merge coplanar neighbors into planar faces, pull vertices onto the planes of their faces. ASCII and binary (binary detected by the size 84 + 50·n, also with a "solid" header).



`examples/spec/stl_import.mcad`: the same angle bracket as B-rep with bore (blue) and mirrored as mesh (orange)



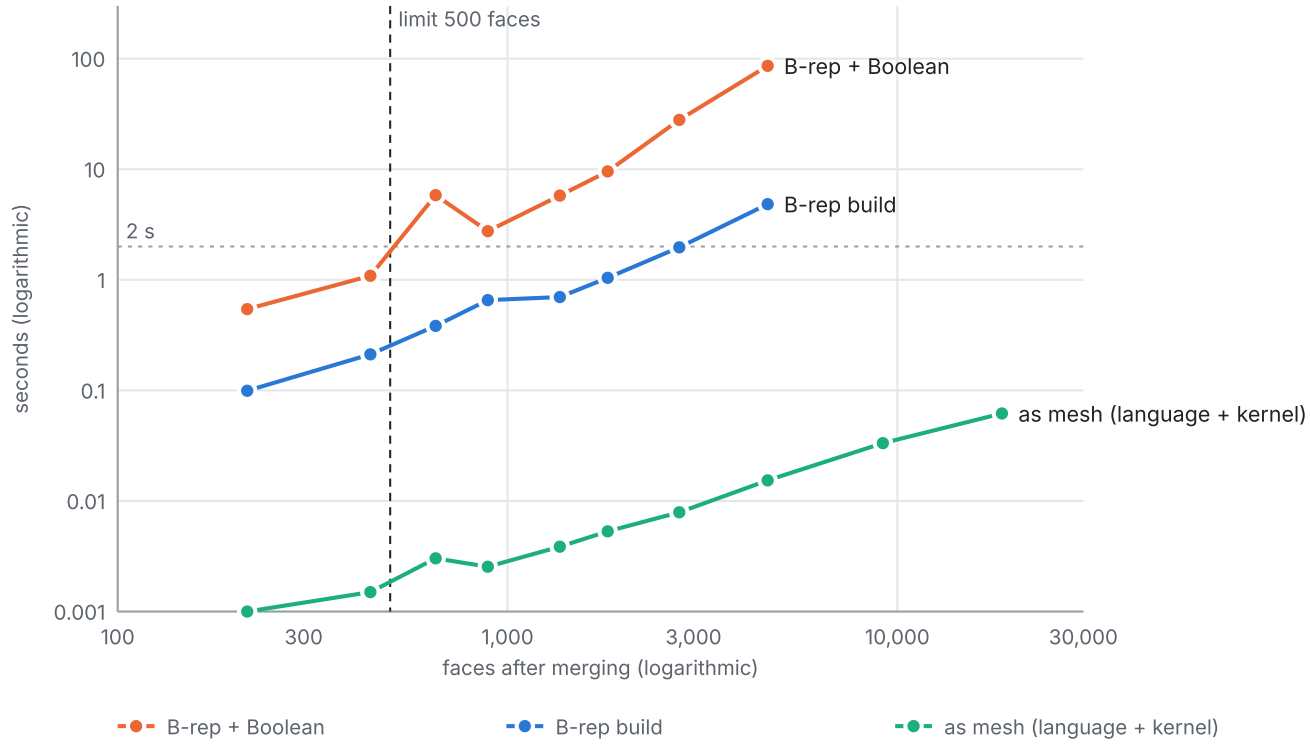
Sphere STL (968 triangles, 890 faces) with `as = "solid"` minus cylinder



Sphere with 20 000 triangles automatically as mesh (edges only from 20°)

```
$ mcad inspect stl_import.mcad --line 8
Z.8: import("winkel.stl");
      Hinweis: B-Rep: 8 Flächen aus 20 Dreiecken
$ mcad inspect stl_import.mcad --line 13
Z.13: translate([-5, 0, 0]) mirror([1, 0, 0]) color("orange") import("winkel.stl", as = "mesh");
      Hinweis: Netz: 20 Dreiecke, 8 Flächen
```

Run time and the 500-face limit



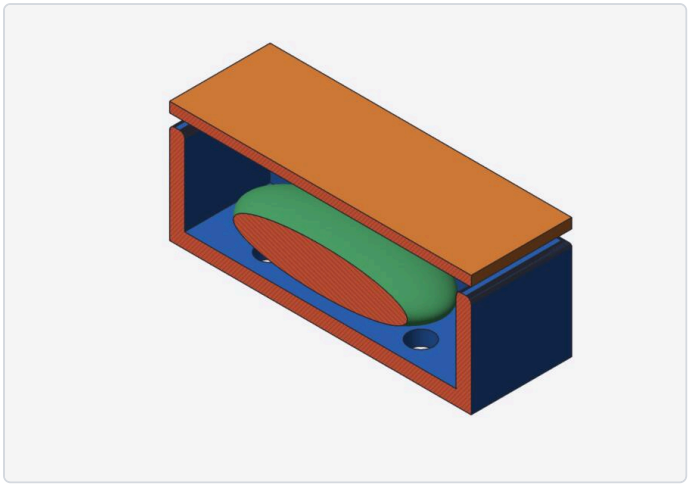
Kernel time of the STL import (UV spheres r 20, median of 3 runs, from 2500 faces 1 run) over the face count: B-rep build (`as = "solid"`), the same body with one Boolean (cylinder through both poles) and as mesh (parsing, preparation and kernel together). Above 5000 triangles measured as mesh only. Values in the table below ("language" = parsing and preparing the STL, "display" = display mesh and image).

triangles	faces	automatic	language	B-rep build	B-rep + Boolean	mesh: kernel / Boolean	display + image
242	215	B-rep	1 ms	0.10 s	0.5 s	0 / 8 ms	128 ms
480	445	B-rep	1 ms	0.21 s	1.1 s	0 / 8 ms	135 ms
722	654	mesh	3 ms	0.38 s	5.8 s	0 / 8 ms	178 ms
968	890	mesh	3 ms	0.65 s	2.8 s	0 / 8 ms	147 ms
1,458	1,362	mesh	4 ms	0.70 s	5.8 s	0 / 7 ms	136 ms
1,984	1,807	mesh	5 ms	1.04 s	9.5 s	0 / 8 ms	140 ms
2,964	2,757	mesh	8 ms	1.96 s	28.0 s	0 / 9 ms	151 ms
5,000	4,646	mesh	15 ms	4.83 s	86.2 s	0 / 8 ms	161 ms
9,940	9,177	mesh	33 ms	–	–	0 / 8 ms	202 ms
20,000	18,556	mesh	62 ms	–	–	0 / 9 ms	198 ms

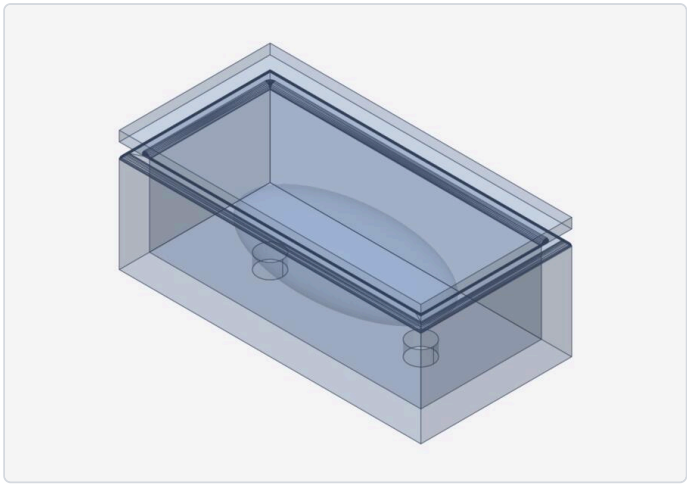
Model (mcad export, re-imported with `as = "solid"`)	triangles	faces	automatic	B-rep build	+ Boolean (cylinder r 6)
Plate 60×40×5 with 8 bores	2,092	533	mesh	0.47 s	1.05 s
Housing from `box.mcad` with fillet	740	582	mesh	0.46 s	0.75 s

Up to 500 faces, import + one Boolean stays at 1.3 s (445 faces). Above that the Boolean time grows superlinearly (890 faces 2.8 s, 4,646 faces 86 s); the times scatter between neighboring sizes because the number of intersection edges at the poles depends on the subdivision of the sphere. As a mesh the kernel costs nothing; parsing and preparing 20,000 triangles takes 62 ms, display mesh and image ≈ 0.2 s regardless of size. Realistic exports lie just above the limit (more faces than a sphere with the same triangle count because of the bores) and automatically become meshes; with `as = "solid"` build and one Boolean take 1–1.5 s.

Combined model: housing from STL, circuit board as mesh



Section y = 10 (`--section-flip`): housing (STL as B-rep, two bores), "circuit board" (sphere with 20 000 triangles, scaled, as mesh) and lid; caps also on the mesh



X-ray: the same model

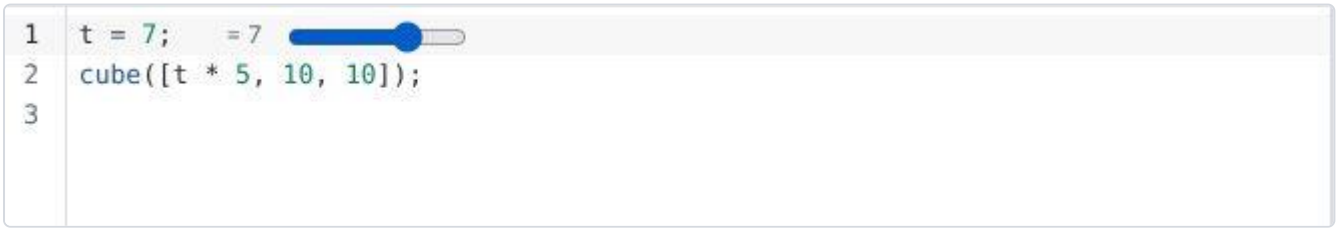
```
$ mcad measure kombi.mcad --volume --interference --clearance platine gehaeuse
kombi.mcad:7:57: Warnung (Auswertung): ungleichmäßige Skalierung verzerrt Verrundungen, Zylinder und
Wandstärken; besser Modulparameter
Volumen all: 21561.7917 mm³
Keine Kollisionen (3 Bauteile, 1 Paare mit überlappender Hülle geprüft, Toleranz 0.001 mm³)
Abstand platine ↔ gehaeuse: 2 mm (± 0.02 mm, Tessellierung 0.01 mm), zwischen [30, 15, 5] und [30, 15, 3]
(Exit 0)
```

Limitations

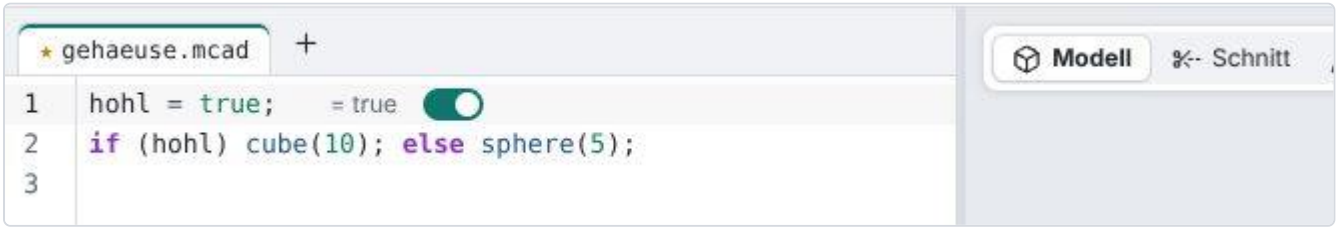
- Collisions with meshes only yes/no (penetration or containment), without intersection volume; distances via the triangles.
- STEP of a mesh is faceted (one plane per face); Booleans and fillets need `as = "solid"` and are slow from a few thousand faces on (curve above).

4 Editor: no inline sliders for numbers

Commit `2d3451e`: the slider at the end of the line of number assignments got in the way of reading and writing and has been removed. Bool literals keep their toggle; numbers can still be changed via the parameter bar and the handles. Playwright B12 checks that no `inline-slider` appears any more.



Before (report 3, B12): `t = 7;` with slider at the end of the line



After (B12, final run): only the toggle on `hohl = true;`

5 Phase C: collision asserts

Language

```
w = 60;   // housing outside
d = 40;
h = 25;
t = 2;    // wall
luft = 1; // minimum clearance to the part

module gehaeuse() difference() {
    cube([w, d, h]);
    translate([t, t, t]) cube([w - 2 * t, d - 2
* t, h]);
}
module deckel() translate([0, 0, h]) cube([w,
d, t]);
module halter() translate([t - 1, d / 2 - 5, t
- 1]) cube([10, 10, 12]);
module kabelfuehrung() translate([w - t - 6, d
/ 2 - 4, t - 1]) cube([7, 8, 6]);

assert_touches() {               // lid rests
flat on the housing
    assert_intersect() {         // holder and
cable guide are joined to the housing
        g: gehaeuse();
        assert_disjoint() {      // holder and
cable guide do not touch
            hl: halter();
            kf: kabelfuehrung();
        }
    }
    k: deckel();
}

assert_keepout(clearance = luft) translate([25,
12, t + 2]) import("../stl/winkel.stl", as =
"mesh");
```

Assert	Violation (warning)
`assert_keepout(clearance = 0)`	a real body intersects the keep-out volume (volume > ε) or is closer than `clearance`; the keep-out volume is shown, but not exported, counted or included in calculations
`assert_disjoint()`	a pair of the children intersects
`assert_intersect()`	the children do not form a connected group of intersecting bodies (touching is not intersecting)
`assert_touches(tol, contact)`	gap > tol, overlap, or only line/point contact with `contact = "face"`

Asserts are groups of their children and can be nested. Intersections exact via Boolean per body pair (volume from the mass properties), contact faces exact via `region2` for lines and arcs, distances via the tessellation with foot points; pairs cached in the engine under content keys.

Test cases vs. formula

`mcad check --json` per case (separate process, cold cache, median of 3 runs). "Check" = time of the assert check, "call" = full CLI call.

Case	Setup	expected	Result	Value	expected value	rel. dev.	check	call
K1	keepout: cuboid protrudes into the plate	intersection	intersection	100.0000 mm³	10·10·1 = 100.0000	$+1.4\cdot10^{-16}$	8.1 ms	77 ms
K2	keepout: clearance 0.5, distance 0.3	too close	too close	0.3000000 mm	3.3 − 3 = 0.3000	$-1.6\cdot10^{-7}$	2.3 ms	72 ms
K3	keepout: clearance 0.5, distance 1	ok	ok	–	distance 1 > 0.5	–	0.0 ms	56 ms
K4	keepout: touching at clearance 0	ok	ok	–	distance 0	–	5.1 ms	68 ms
K5	keepout: sphere r 2 half in the plate	intersection	intersection	16.7551 mm³	$2/3\cdot\pi\cdot r^3$ = 16.7552	$-8.0\cdot10^{-7}$	238.8 ms	313 ms
K6	keepout: STL as mesh in the plate	intersection	intersection	–	mesh: yes/no only	–	2.9 ms	57 ms
K7	keepout: two parts, only b in the keep-out volume	intersection	intersection	125.0000 mm³	5·5·5 = 125.0000	$-2.3\cdot10^{-16}$	8.0 ms	60 ms
D1	disjoint: overlapping	intersection	intersection	200.0000 mm³	2·10·10 = 200.0000	$-1.4\cdot10^{-16}$	10.7 ms	74 ms

Case	Setup	expected	Result	Value	expected value	rel. dev.	check	call
D2	disjoint: touching	ok	ok	–	touching allowed	–	5.7 ms	59 ms
D3	disjoint: cylinder half in the cuboid	intersection	intersection	62.8319 mm³	π·r²·h/2 = 62.8319	−2.8·10⁻⁹	44.5 ms	137 ms
I1	intersect: chain a–b–c	ok	ok	–	connected	–	22.8 ms	95 ms
I2	intersect: fourth child 14 mm away	separated	separated	14.0000 mm	40 − 26 = 14.0000	0	27.3 ms	97 ms
I3	intersect: touching only	separated	separated	0.0000000 mm	touching ≠ intersection = 0.0000	–	8.2 ms	72 ms
T1	touches: face 5×5	ok	ok (face, exact)	25.0000 mm²	5·5 = 25.0000	0	10.7 ms	71 ms
T2	touches: cylinder r 3 on face	ok	ok (face, exact)	28.2743 mm²	π·r² = 28.2743	−1.3·10⁻¹⁶	34.9 ms	115 ms
T3	touches: lid on housing rim	ok	ok (face, exact)	384.0000 mm²	60·40 − 56·36 = 384.0000	0	17.4 ms	89 ms
T4	touches: lying cylinder (line)	line contact only	line contact only	–	line contact	–	23.9 ms	89 ms
T5	touches: the same, contact = "any"	ok	ok (line, exact)	–	line allowed	–	24.6 ms	120 ms
T6	touches: sphere on plate (point)	point contact only	point contact only	–	point contact	–	176.4 ms	250 ms
T7	touches: gap 0.2	gap	gap	0.1999998 mm	10.2 − 10 = 0.2000	−9.5·10⁻⁷	1.6 ms	57 ms
T8	touches: gap 0.2 at tol 0.3	ok	ok (face, exact)	–	within tol	–	3.6 ms	53 ms
T9	touches: overlap 1 mm	intersection	intersection	100.0000 mm³	1·10·10 = 100.0000	+1.4·10⁻¹⁶	11.4 ms	68 ms
T10	touches: cuboids edge to edge	line contact only	line contact only	–	edge to edge	–	6.2 ms	59 ms

Distances (K2, T7) come from the f32 tessellation and deviate by ≈ 10^{-7} relative; volumes and contact areas are exact. The sphere (K5, T6) costs ≈ 0.2 s for the Boolean with the spherical surface.

Export, CLI images

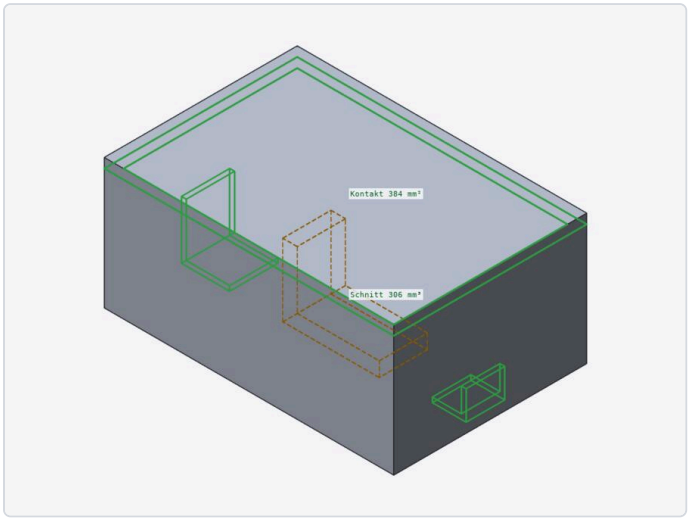
```
$ mcad check asserts.mcad
Asserts: 4 von 4 geprüft, 0 verletzt
asserts.mcad: 0 Fehler, 0 Warnungen
(Exit 0)

$ mcad check asserts_bad.mcad
asserts_bad.mcad:21:1: Warnung (Kernel): assert_touches: g + hl + kf und k überschneiden sich (Schnittvolumen
100 mm³)
asserts_bad.mcad:33:1: Warnung (Kernel): assert_keepout: g hält den Abstand 3 mm zum Sperrvolumen nicht ein: 2
mm (± 0.0305) von [42.4444, 12, 2] bis [42.4444, 12, 4] [Zeile 14: box (Instanz g: gehaeuse)]
Asserts: 4 von 4 geprüft, 2 verletzt (2 Verletzung(en); mcad export bricht dann ab, außer mit --ignore-asserts)
asserts_bad.mcad: 0 Fehler, 2 Warnungen
(Exit 0)

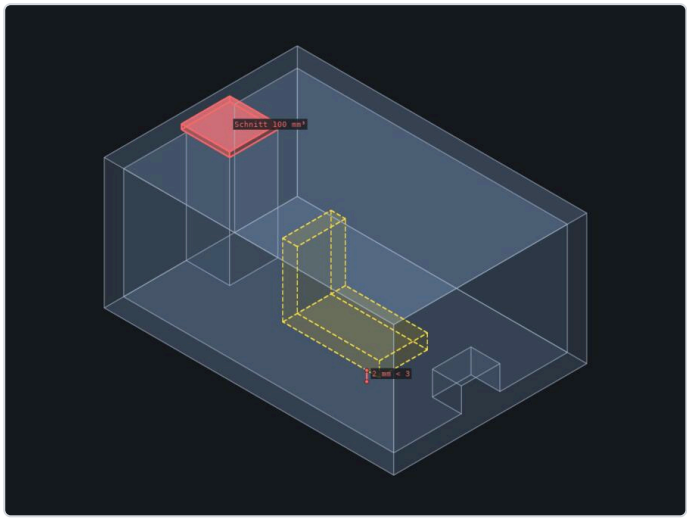
$ mcad export asserts_bad.mcad -o x.stl
asserts_bad.mcad:21:1: Warnung (Kernel): assert_touches: g + hl + kf und k überschneiden sich (Schnittvolumen
100 mm³)
asserts_bad.mcad:33:1: Warnung (Kernel): assert_keepout: g hält den Abstand 3 mm zum Sperrvolumen nicht ein: 2
mm (± 0.0305) von [42.4444, 12, 2] bis [42.4444, 12, 4] [Zeile 14: box (Instanz g: gehaeuse)]
Export abgebrochen: 2 Assert-Verletzung(en), nichts geschrieben:
  Z.21 assert_touches: g + hl + kf und k überschneiden sich (Schnittvolumen 100 mm³)
  Z.33 assert_keepout: g hält den Abstand 3 mm zum Sperrvolumen nicht ein: 2 mm (± 0.0305) von [42.4444, 12, 2]
bis [42.4444, 12, 4]
Trotzdem exportieren: --ignore-asserts
(Exit 4)

$ mcad export asserts_bad.mcad -o x.stl --ignore-asserts
```

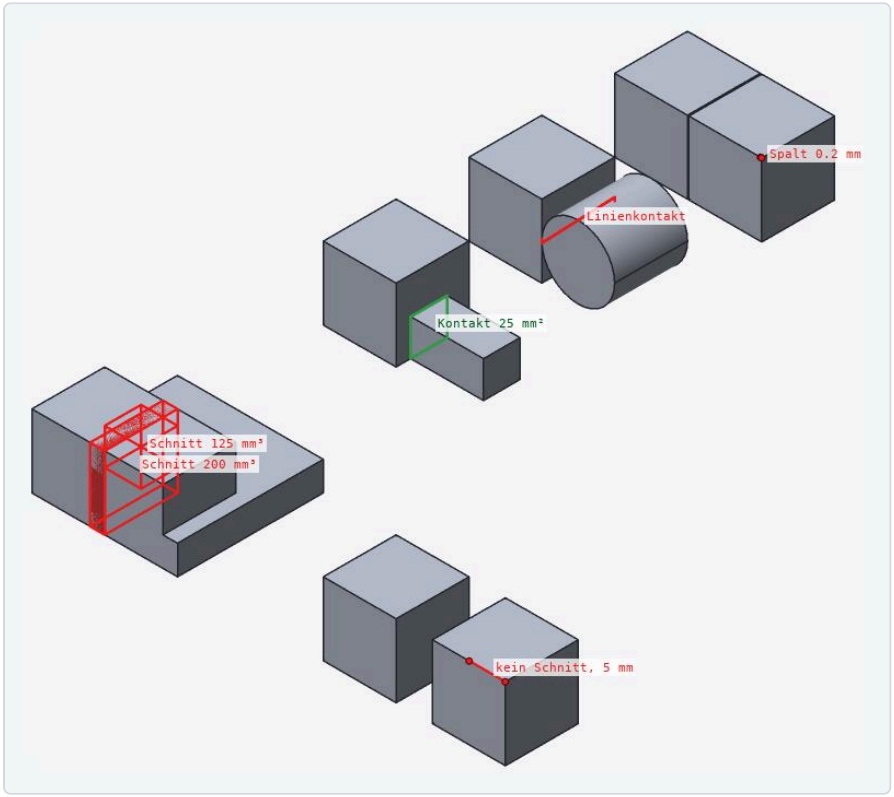
```
asserts_bad.mcad:21:1: Warnung (Kernel): assert_touches: g + hl + kf und k überschneiden sich (Schnittvolumen
100 mm³)
asserts_bad.mcad:33:1: Warnung (Kernel): assert_keepout: g hält den Abstand 3 mm zum Sperrvolumen nicht ein: 2
mm (± 0.0305) von [42.4444, 12, 2] bis [42.4444, 12, 4] [Zeile 14: box (Instanz g: gehaeuse)]
Warnung: trotz 2 Assert-Verletzung(en) exportiert (--ignore-asserts)
x.stl: 56 Dreiecke, 2884 Bytes, Volumen 20742 mm³
(Exit 0)
```



`mcad render asserts.mcad --show asserts`: contact lid/housing 384 mm² green, intended intersection holder/cable guide green, keep-out volume yellow dashed



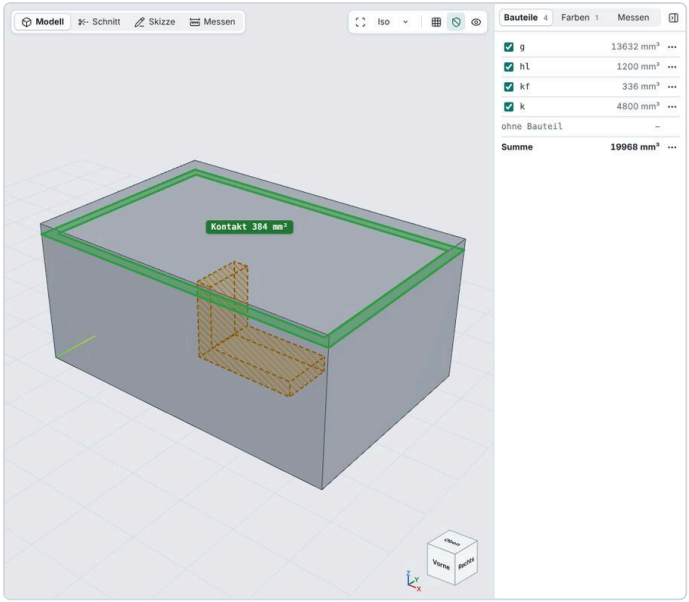
Violation (`luft = 3`, holder too high), `--xray --theme dark`: intersection red, distance "2 mm < 3" as a line with foot points



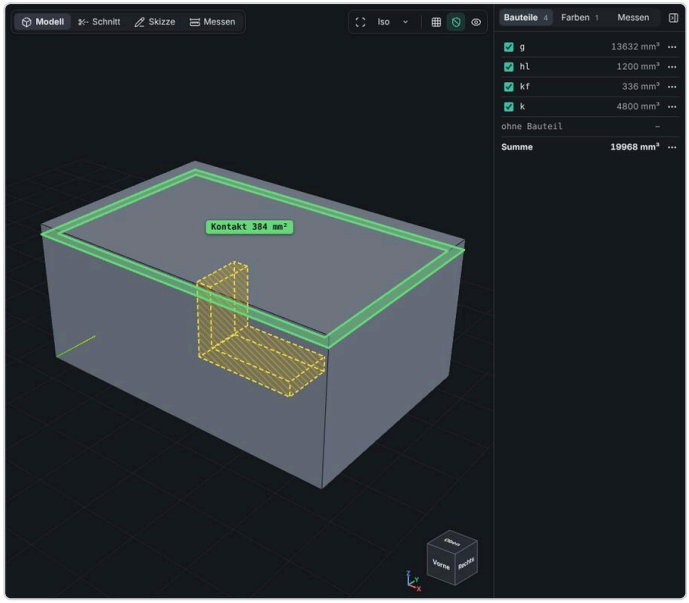
`faelle.mcad --show asserts`: all kinds side by side – keepout (intersection), disjoint (intersection 200 mm³), intersect (separated), face contact 25 mm², line contact, gap 0.2 mm

UI (browser, Wasm kernel)

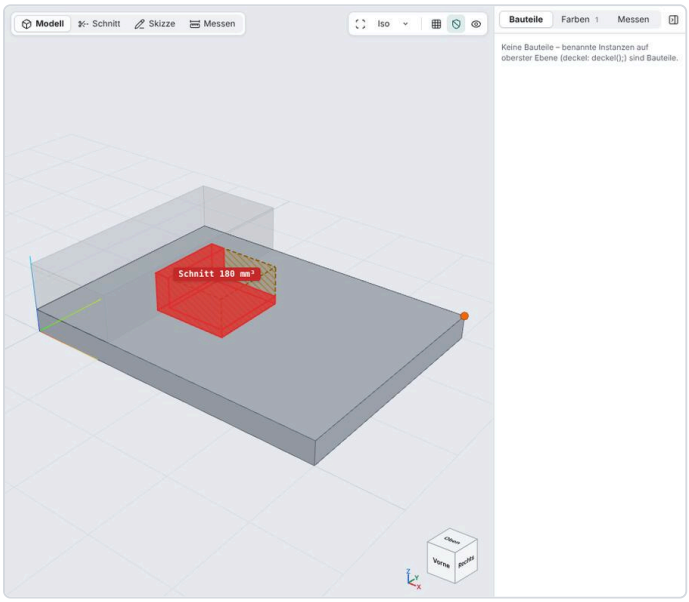
Keep-out volume hatched in yellow and visible through the model, not selectable or measurable, toggle "Keep-out volume" (German UI: „Sperrvolumen“, remembered); intersection body of a violation red, contact green, gap as dimension line; status bar "n assert violations", a click on it or on the warning at the end of the line frames the violation; every export asks for confirmation when there is a violation. Images from the final Playwright run (C01–C05).



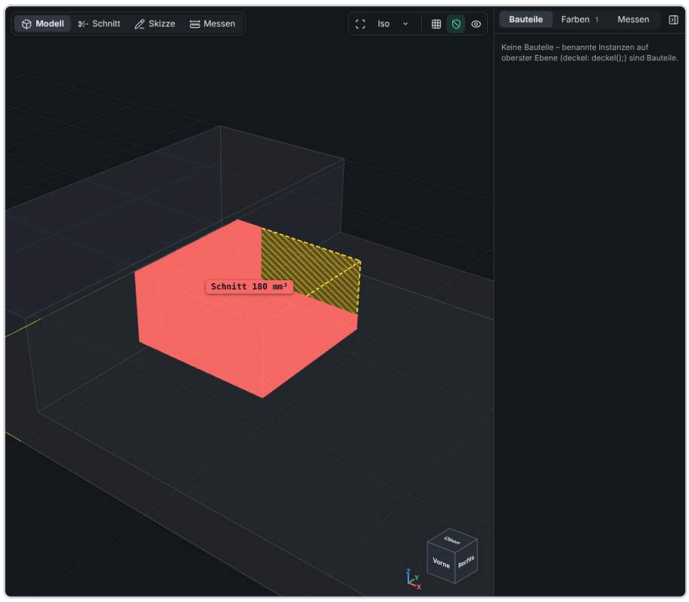
C01 light: keep-out volume (STL angle bracket) hatched, contact 384 mm²



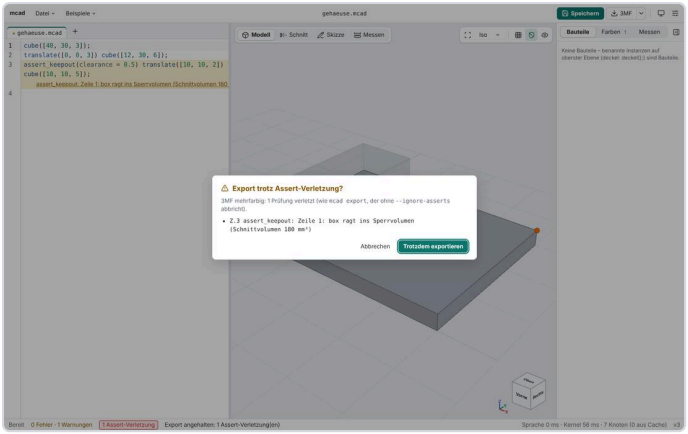
C01 dark



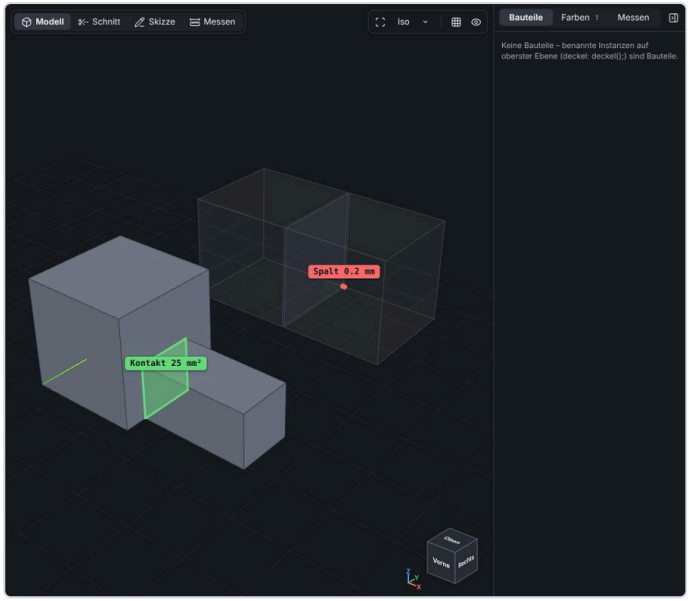
C02: violation, intersection body red with "intersection 180 mm³" (German UI: „Schnitt 180 mm³“)



C02 dark: framed after clicking "1 assert violation"



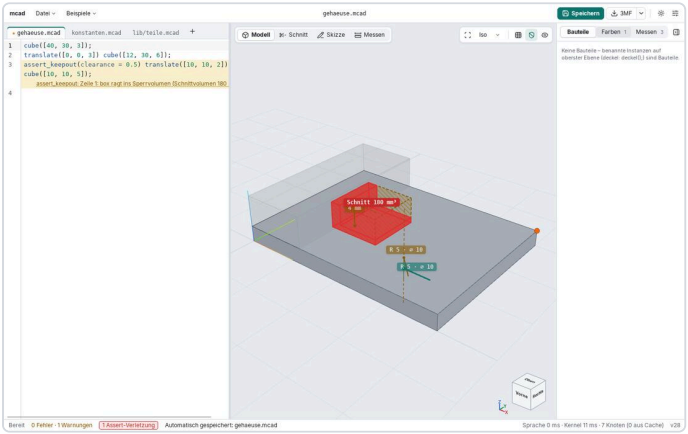
C04: "Export despite assert violation?" (German UI: „Export trotz Assert-Verletzung?“) with list



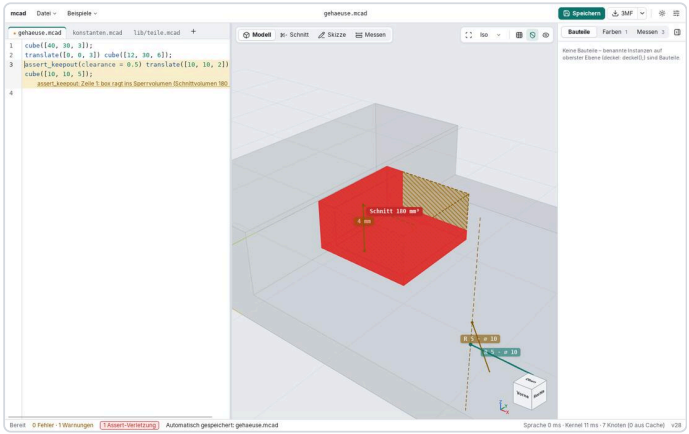
C05 dark: contact area 25 mm² green, gap 0.2 mm with dimension line

Desktop app (Tauri T19)

T19 passed (1.5 s): intersection volume 180 mm³ (10·10·1 + 2·10·4 = 180); keep-out volume drawn; intersection body drawn in red; violation focused; confirmation text: „Z.3 assert_keepout: Zeile 1: box ragt ins Sperrvolumen (Schnittvolumen 180 mm³)“.



T19: keep-out volume with violation (Tauri, native kernel)



T19: click in the status bar: intersection body framed

Wasm build

emsdk 6.0.13 and wasm-bindgen-cli 0.2.129 could be set up in the container (downloads via the proxy); `smlib.wasm` rebuilt (≈ 5 min, 244 targets, with `sc_offset` and the contact locations from phase D), `cad-wasm` ≈ 65 s. Sizes after `wasm-opt -O3`:

File	before phase A (852fbc1)	after phase C	now	Δ vs. before	gzip -9 before → now
`cad_wasm_bg.wasm`	4,545,057	4,887,251	4,923,698	+8.3 %	1,925,346 → 2,093,609
`smlib.wasm`	5,283,311	5,289,122	5,305,410	+0.4 %	1,862,076 → 1,874,500

Values before phase A and after phase C from the phase C material (same tools, old state as a separate build); "now" = `web/src/wasm-pkg` of the final state.

Limitations

- Collision with meshes only yes/no, without volume (K6).
- The keepout check computes Booleans in the worker: on a hole plate 120×80 with 24 bores and fillet ≈ 2.5 s the first time (afterwards from the cache); the worker is busy for that long.

- Point contact (sphere on plate) is detected, but reported and drawn without location.

6 Phase D: user feedback

For each item the same call with the release build before phase D (`cad` 0.2.1) and with the final state (`mcad` 0.3.0); inputs as `.scad`, because the old build does not know `.mcad`.

No.	Item	before	after
D1	Kernel errors with location The message names both operands with line and the location; JSON `at` and `nodes`; render and UI mark the location.	`$ cad check dom_rippe.scad` `dom_rippe.scad:8:13: Warnung (Auswertung): unbekanntes Modul `slot`` `dom_rippe.scad:1:1: Fehler (Kernel): Vereinigung gescheitert: the result is not a closed solid: the operands touch tangentially or along an edge there (let them overlap a little or keep a gap)` `dom_rippe.scad: 1 Fehler, 1 Warnungen` `(Exit 1)` `$ cad check f09a_absturz_tangentialer_dom.scad` `f09a_absturz_tangentialer_dom.scad:7:1: Fehler (Kernel): Vereinigung gescheitert: the result is not a closed solid: the operands touch tangentially or along an edge there (let them overlap a little or keep a gap)` `f09a_absturz_tangentialer_dom.scad: 1 Fehler, 0 Warnungen` `(Exit 1)` `$ cad check f08c_differenz_torus_tangential.scad` `f08c_differenz_torus_tangential.scad:7:1: Fehler (Kernel): Differenz gescheitert: the result is not a closed solid: the operands touch tangentially or along an edge there (let them overlap a little or keep a gap)` `f08c_differenz_torus_tangential.scad: 1 Fehler, 0 Warnungen` `(Exit 1)`	`$ mcad check dom_rippe.scad` `dom_rippe.scad:1:1: Fehler (Kernel): Vereinigung gescheitert: Zeile 15 (dom) und Zeile 16 (rippe) berühren sich tangential bei [7, 15, 5]: the result is not a closed solid: the operands touch tangentially or along an edge there (let them overlap a little or keep a gap)` `dom_rippe.scad: 1 Fehler, 0 Warnungen` `(Exit 1)` `$ mcad check f09a_absturz_tangentialer_dom.scad` `f09a_absturz_tangentialer_dom.scad:7:1: Fehler (Kernel): Vereinigung gescheitert: Zeile 9 (box) und Zeile 10 (cylinder) berühren sich tangential bei [2, 10, 6]: the result is not a closed solid: the operands touch tangentially or along an edge there (let them overlap a little or keep a gap)` `f09a_absturz_tangentialer_dom.scad: 1 Fehler, 0 Warnungen` `(Exit 1)` `$ mcad check f08c_differenz_torus_tangential.scad` `f08c_differenz_torus_tangential.scad:7:1: Fehler (Kernel): Differenz gescheitert: Zeile 8 (cylinder) und Zeile 9 (torus) berühren sich tangential bei [-20, 0, -1] (und 1 weitere Stelle): the result is not a closed solid: the operands touch tangentially or along an edge there (let them overlap a little or keep a gap)` `f08c_differenz_torus_tangential.scad: 1 Fehler, 0 Warnungen` `(Exit 1)`

D2	`-D` without module context
	Sets top-level assignments like `openscad -D`; unknown name → warning.
		`$ cad measure teil.scad -D w=40 --volume all`
		`cad measure: -D braucht einen Modulkontext (--module oder --line in einem Modul)`
		`(Exit 2)`
		`$ cad check teil.scad -D breite=3`
		`cad check: -D braucht einen Modulkontext (--module oder --line in einem Modul)`
		`(Exit 2)`
			`$ mcad measure teil.scad -D w=40 --volume all`
			`Volumen all: 6400 mm³`
			`(Exit 0)`
			`$ mcad check teil.scad -D breite=3`
			`teil.scad:1:1: Warnung (Auswertung): -D breite: keine Zuweisung `breite` auf oberster Ebene der Hauptdatei (ignoriert)`
			`teil.scad: 0 Fehler, 1 Warnungen`
			`(Exit 0)`

D3	STEP with separate bodies
	One named body per part or top-level object; `--merge` writes one body as before.
		`$ cad export pair.scad -o pair.step`
		`pair.step: 1 Körper, 6 Flächen, 12 Kanten, 6703 Bytes, Volumen 2000 mm³`
		`(Exit 0)`
		`# STEP: 1 MANIFOLD_SOLID_BREP ['pair 1']`
			`$ mcad export pair.scad -o pair.step`
			`pair.step: 2 Körper (a, b), 12 Flächen, 24 Kanten, 11847 Bytes, Volumen 2000 mm³`
			`(Exit 0)`
			`# STEP: 2 MANIFOLD_SOLID_BREP ['a', 'b']`

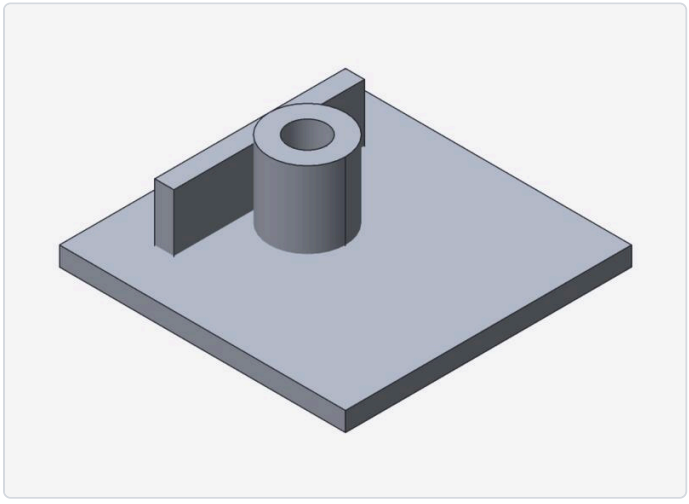
No.	Item	before	after

D5	`slot(p1, p2, r)` Slot as exact 2D primitive (before: unknown module, plate without hole).	`$ cad measure slot.scad --volume all --section z=2`
		`slot.scad:3:45: Warnung (Auswertung): unbekanntes Modul `slot``
		`slot.scad:3:27: Warnung (Kernel): kein Profil`
		`Schnitt z = 2 (entfernt z > 2): 1 Bereich(e), Fläche 800 mm² (exakt: Konturen aus Geraden und Kreisen der B-Rep)`
		`  Bereich 1: 800 mm², Hülle [0, 0, 2] – [40, 20, 2], 0 Loch/Löcher`
		`  kleinste Wandstärke ≈ 20 mm von [37.5, 20, 2] bis [37.5, 0, 2] (8 Stellen je Konturabschnitt; genau: --wall P)`
		`Volumen all: 3200 mm³`
		`(Exit 0)`
			`$ mcad measure slot.scad --volume all --section z=2`
			`Schnitt z = 2 (entfernt z > 2): 1 Bereich(e), Fläche 651.7257 mm² (exakt: Konturen aus Geraden und Kreisen der B-Rep)`
			`  Bereich 1: 651.7257 mm², Hülle [0, 0, 2] – [40, 20, 2], 1 Loch/Löcher`
			`  Bogen R 3 · ⌀ 6 bei [10, 10, 2] (center@[10,10,2])`
			`  Bogen R 3 · ⌀ 6 bei [30, 10, 2] (center@[30,10,2])`
			`  kleinste Wandstärke ≈ 7 mm von [27.5, 20, 2] bis [27.5, 13, 2] (8 Stellen je Konturabschnitt; genau: --wall P)`
			`Volumen all: 2606.9027 mm³`
			`(Exit 0)`

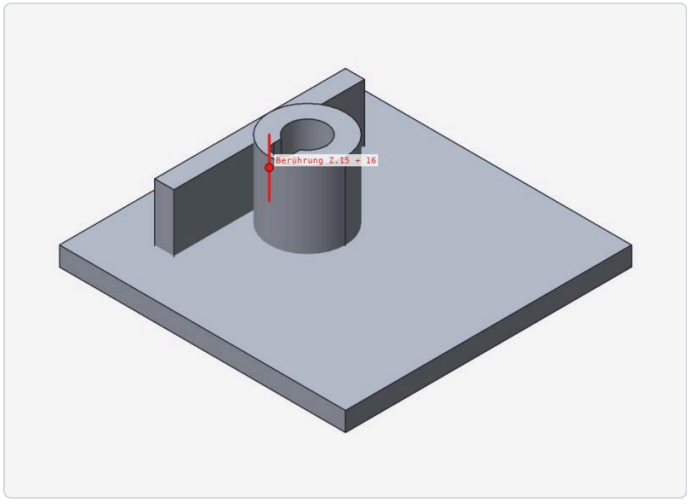
D7	`-m 'modul(args)'` Module context with arguments as in the code, nested `teile.schraube(d = 3)`; errors with exit 2.	`$ cad export halter.scad -m 'halter(30, h = 5)' -o halter.stl`
		`cad export: kein Ergebnis`
		`(Exit 3)`
		`$ cad export halter.scad -m 'teile.schraube(d = 3)' -o schraube.stl`
		`cad export: kein Ergebnis`
		`(Exit 3)`
		`$ cad check halter.scad -m 'halter(x = 1)'`
		`halter.scad:1:1: Fehler (Auswertung): Modul `halter(x = 1)` nicht gefunden`
		`halter.scad: 1 Fehler, 0 Warnungen`
		`(Exit 1)`
			`$ mcad export halter.scad -m 'halter(30, h = 5)' -o halter.stl`
			`halter.stl: 12 Dreiecke, 684 Bytes, Volumen 750 mm³`
			`(Exit 0)`
			`$ mcad export halter.scad -m 'teile.schraube(d = 3)' -o schraube.stl`
			`schraube.stl: 188 Dreiecke, 9484 Bytes, Volumen 70.6858 mm³`
			`(Exit 0)`
			`$ mcad check halter.scad -m 'halter(x = 1)'`
			`mcad check: -m 'halter(x = 1)': Modul `halter` hat keinen Parameter `x` (Parameter: w, h, d)`
			`(Exit 2)`

D1 Kernel errors with location

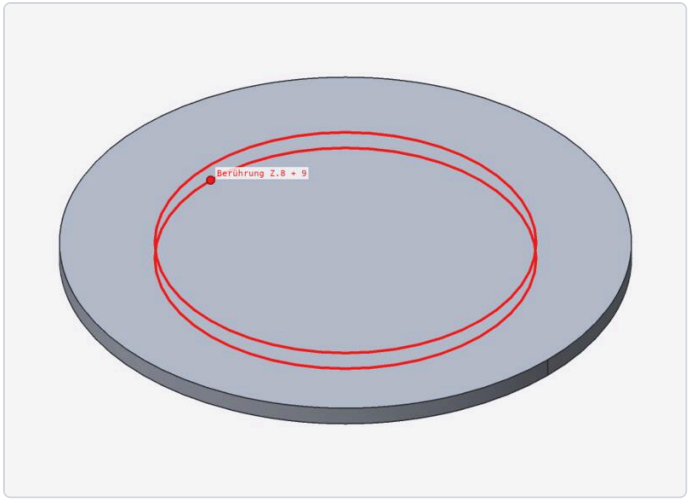
If a Boolean result is not a closed solid, the facade (version 6) returns the non-manifold edges as polylines; the engine knows the node per operand and names the touching operands. JSON diagnostic with `at` (context coordinates) and `nodes`; `mcad render` marks every location (edges red, point, label „Berührung Z.15 + 16“ = contact lines 15 + 16), the UI does the same, and a click on the message frames the location.



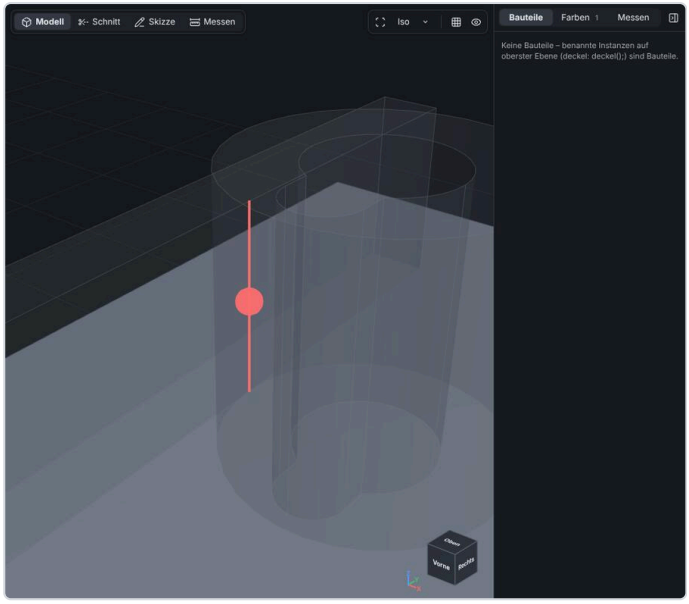
Before: `cad render dom_rippe.scad` without marking (and without `slot`)



After: contact line red with „Berührung Z.15 + 16“



`f08c_differenz_torus_tangential`: cylinder – torus, both contact circles red



UI (Playwright D01, dark): a click on the message frames the location

D3 STEP: body per part and object

Default: one `MANIFOLD_SOLID_BREP` per part (name of the part) and per top-level object outside the parts (name of the object through transformations); touching parts stay separate (before: merged). If objects overlap (sum of volumes > volume of the union), the export prints a notice. `--merge` writes the root as one body as before. Of 64 example models, 12 changed their body count (state `8ac92d6`, read with OCCT):

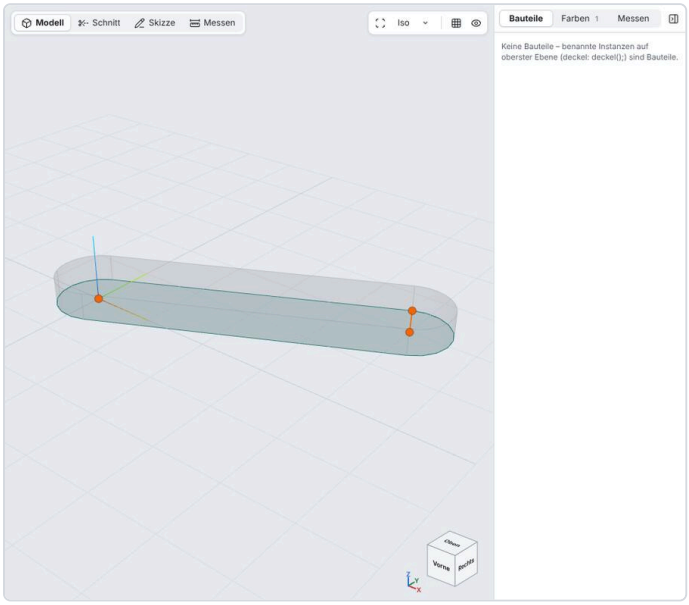
Model	before (`--merge`)	after	Names after
`openscad-compat/07_crown_ring.scad`	1	17	difference, group 1, group 2, group 3, group 4, group 5, group 6, group 7, group
`openscad-compat/14_pipe_bend.scad`	1	3	rotate_extrude, linear_extrude, linear_extrude 2
`openscad-compat/25_root_modifier.scad`	1	2	box, cylinder
`openscad-compat/30_each_ranges.scad`	1	8	box, group 1, group 2, group 3, group 4, group 5, group 6, group 7
`spec/asserts.scad`	1	4	g, hl, kf, k
`spec/gehaeuse.scad`	2	3	k, deckel, deckel 2
`spec/kabelfuehrung.scad`	2	4	sweep, sweep 2, group 1, group 2
`spec/knauf_mirror.scad`	2	3	k, knauf, cylinder
`spec/skizze_deckel.scad`	1	2	difference, linear_extrude
`spec/skizze_kabelkanal.scad`	1	4	linear_extrude, linear_extrude 2 1, linear_extrude 2 2, linear_extrude 3
`spec/text_schild.scad`	1	41	linear_extrude, linear_extrude 2, linear_extrude 3 1, linear_extrude 3 2, linear
`spec/winkel_anker.scad`	3	6	a, b, schraube, schraube 2, schraube 3, box

With the notice "objects overlap" (German: „Objekte überschneiden sich“; final run `verify.py --bodies`, volumes not compared there): `07_crown_ring` (17 bodies, sum 2,239.3 mm³ vs. 2,161.0 mm³ of the union), `asserts` (4 bodies, sum 19,968.0 mm³ vs. 19,662.0 mm³ of the union), `kabelfuehrung` (4 bodies, sum 7,755.5 mm³ vs. 7,748.1 mm³ of the union), `knauf_mirror` (3 bodies, sum 15,685.7 mm³ vs. 15,681.7 mm³ of the union), `skizze_kabelkanal` (4 bodies, sum 10,988.2 mm³ vs. 9,898.7 mm³ of the union), `winkel_anker` (6 bodies, sum 8,401.4 mm³ vs. 8,337.8 mm³ of the union) – see limitations.

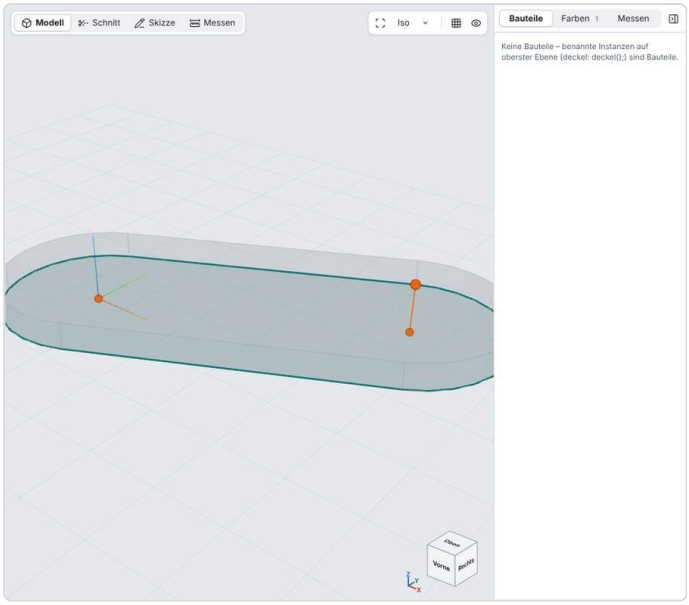
D5 `slot(p1, p2, r)`

Call, extruded by 1	mcad	$\pi r^2 + 2r\cdot\lvert p2 - p1\rvert$	rel. dev.
`slot([0, 0], [20, 10], 3)`	162.4384	162.4384	$+7.7\cdot10^{-8}$
`slot([-5, 1], [7, 1], 2)`	60.5664	60.5664	$+4.9\cdot10^{-7}$
`slot([5, 5], [5, 5], 2)`	12.5664	12.5664	$+2.3\cdot10^{-6}$
`slot([0, 0], [100, 0], 0.5)`	100.7854	100.7854	$+1.8\cdot10^{-8}$

Area from the volume integration (state `91f8ef6`); the contour itself consists of two semicircles and two lines (exact arcs in the section, see table above). `p1 == p2` yields a circle; points also as grid points/anchors, `d = …` instead of `r`; handles for p1, p2 and r.



UI (D02): handles p1, p2 and r on `linear_extrude(2) slot(…)`



D02: radius dragged, the code follows

D6 JSON schema 2

	before	after
`mcad inspect fillet_zonen --line 12 --json`	`schema_version 1`, `selected_edges: 2` (number)	`schema_version 2`, `selected_edges`: list of 2 polylines with 36 points (world coordinates), `selected_edge_count: 2`
`outline`	grid plane `axis` as name, literals/parts inconsistent	`axis` 0|1|2 as in the scene (+ `axis_name`), `variables[].literal` as range|null, `instances[].part` as name|null

7 Phase E: rename `cad` → `mcad`

Binary, program name, messages and manual are called `mcad` (version 0.3.0); new files get `.mcad`, and `.mcad` and `.scad` open everywhere (CLI, `include`/`use`, dialogs, folder projects, upload, ZIP, examples menu). Inventory via `git grep` before the rename: 1703 hits for `-w cad` outside the reports, 187 `.scad` files.

Area	before	after
Binary, messages, manual, skill	`cad`, `cad check: …`	`mcad`, `mcad check: …`; test `usage::no_old_program_name`
Environment, configuration, socket	`CAD_*`, `~/.config/controller-cad/`, `$XDG_RUNTIME_DIR/controller-cad`	`MCAD_*` (old names as fallback), `~/.config/mcad/` (the old file applies if the new one is missing), `$XDG_RUNTIME_DIR/mcad`
Desktop app	productName `controller-cad`, app ID `dev.controller-cad.desktop`	`mcad`, `dev.mcad.desktop`, window title "… – mcad"; list of recently opened files migrated once

Area	before	after
File types, package	desktop file with `application/x-openscad`, Arch package `controller-cad`	`text/x-mcad` and `application/x-openscad` (`packaging/linux/mcad.xml`), package `mcad` with `replaces=('controller-cad')`
Examples, test data	94 files `*.scad`	`*.mcad` via `git mv`, references updated
STEP/3MF metadata, web title	`controller-cad`, „Parametrisches CAD“	`mcad`, „mcad – Parametrisches CAD“

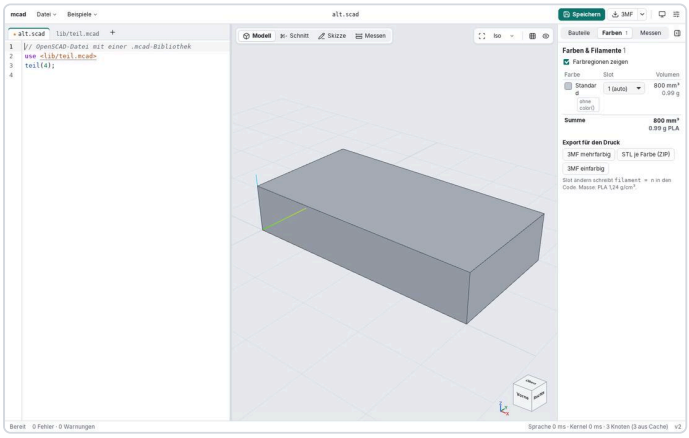
unchanged	Reason
Crate names `cad-*`, skill folder `cad-cli`	internal; paths in briefs and tools
Repository `controller-cad`, Docker image/volume of the Arch build environment	name of the repository and of the build infrastructure
IndexedDB name `controller-cad` (browser)	holds the users' saved projects
Font name "cad stencil mono"	appears in models
`examples/openscad-compat/*.scad`, `tests/openscad-compat`, `tests/farben-pruefung`, case study report 8	are compared with OpenSCAD or are snapshots; incidentally check that `.scad` still works
Reports `docs/reports/**`	snapshots

```
$ mcad --version
mcad 0.3.0

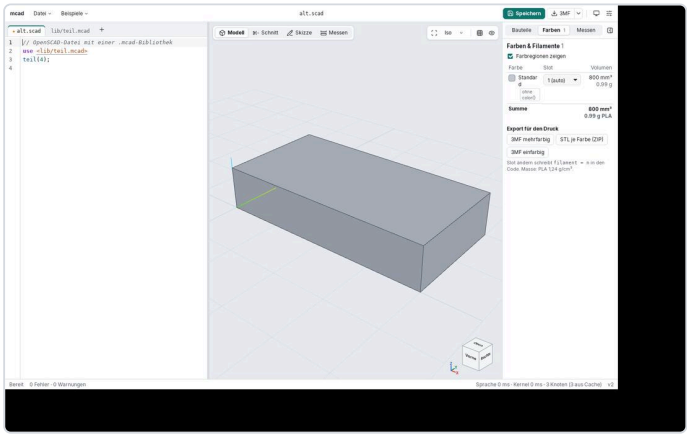
$ mcad check teil.mcad
teil.mcad: 0 Fehler, 0 Warnungen
(Exit 0)

$ mcad measure lib/halter.scad -m 'halter(30)' --volume all
Volumen all: 300 mm³
(Exit 0)
```

Tauri T20 **passed**: file alt.scad; no diagnostics ([]); project files ["alt.scad","lib/teil.mcad"]; header bar "mcad"; window title "alt.scad – mcad"; WM_CLASS dev.mcad.desktop/Dev.mcad.desktop.



T20: alt.scad with use <lib/teil.mcad>, header bar "mcad"



T20: window "alt.scad – mcad", WM_CLASS Dev.mcad.desktop (X screenshot)

8 Test runs on the final state

Command	Result	Duration
`cargo fmt --all --check`	clean	2 s
`cargo test --workspace --no-fail-fast` (with Tauri/WebKitGTK, `CARGO_INCREMENTAL=0`)	658 passed, 0 failed, 0 ignored (106 test binaries)	6.4 min

Command	Result	Duration
`cargo clippy --workspace --all-targets -- -D warnings` and `-p cad-wasm --target wasm32-unknown-unknown`	clean	40 s + 19 s
`cargo check -p kernel-solidcore --features solidcore`	ok	3 s
`cd web && pnpm test` (Vitest)	246 passed, 0 failed	3 s
`pnpm typecheck`	clean	3 s
`pnpm e2e` (Playwright 1.56.1, Chromium with SwiftShader, full suite, JSON reporter)	149 passed, 0 failed (17 files)	20.5 min
`node web/e2e-tauri/desktop.e2e.mjs --bin target/release/mcad` (tauri-driver, Xvfb)	20 / 20 passed	74 s
`MCAD=target/release/mcad python3 tests/openscad-compat/compare.py --eval`	36 / 36	18 s
`tests/step/verify.py` (OCCT, cadquery-ocp 8.0.1.1, `--merge` against the root)	64 / 64	1.8 min
`tests/step/verify.py --bodies` (new default: body per part/object)	64 / 64	86 s
`tests/alltag/check.sh -j 4` (everyday models: check, parts, export)	19 models, 0 with problems, 97 s, exports in /tmp/alltag-sbggskb9 (deleted)	1.6 min

Found and fixed: `tests/step/verify.py` aborted on the final state with `SyntaxError: unmatched ')'` before checking a single file – when the default was switched from `--cad` to `MCAD` (commit `4db585b`), an extra parenthesis had been left behind. Fixed in `0ea0d78`; afterwards 64/64 in both modes. Otherwise all runs were green on the first pass; nothing skipped or disabled.

All runs sequentially on the same machine (4 cores), logs in `reports/tmp/report21/final/`. New in these phases: Playwright C01, C02, C03, C04, C05, C06, C07, D01, D02, B12; Tauri T19, T20.

Tauri E2E

No.	Scenario	Result	Duration
T01	App starts with a file, scene appears (native kernel)	passed	2.9 s
T02	External change without unsaved changes: silently reloaded, undo as a separate step	passed	0.9 s
T03	Git markers in the gutter (new, changed, deleted)	passed	1.1 s
T04	External change with unsaved changes: notice bar, compare, keep mine, save without loop	passed	2.8 s
T05	Deleted and renamed file	passed	1.4 s
T06	Agent flow: change file via shell, `mcad check`, UI follows (image sequence)	passed	1.2 s
T07	`mcad ui status` / `show` / `screenshot` against the running app	passed	1.1 s
T08	Section view in Tauri (cap, contour, dragging, CLI call)	passed	6.3 s
T09	Watchdog in Tauri: hanging kernel → engine process replaced, node blocked	passed	7.7 s
T10	Folder as project, autosave	passed	3.8 s
T11	Keystroke → scene natively (measurement)	passed	14.1 s
T12	Colors & filaments in Tauri: list, change slot, 3MF multicolor (native kernel)	passed	3.6 s
T13	Dark theme in Tauri: contrast (computed styles), view, editor	passed	0.8 s
T15	Report 12 in Tauri: color regions per instance, filament() table, object "without part", 3MF without color group	passed	0.9 s
T16	Sketch editor in Tauri: drawing mode, rectangle on grid points, snap indicator, undo (TauriTransport, native kernel)	passed	2.2 s

No.	Scenario	Result	Duration
T17	Measuring in Tauri: radius, distance of parallel faces, radius in the section (TauriTransport, native kernel)	passed	2.6 s
T18	UI rework in Tauri: bars without wrapping at 1280/1440/1920 (light/dark), menus, view cube, side panel	passed	10.6 s
T19	Asserts in Tauri: keep-out volume, framing a violation, export confirmation (native kernel)	passed	1.5 s
T14	Theme "System" in Tauri follows the GTK color scheme (prefers-color-scheme)	passed	4.5 s
T20	Rename to mcad: window title, app ID (WM_CLASS), header bar; .scad opens with use <....mcad>	passed	2.9 s

9 Limitations and open items

Item	Cause or status
Offset/`minkowski` with a sphere at non-right-angled corners	SMLib approximates the corner spheres with trim curves up to 0.33 mm off the edges (tetrahedron) or fails (wedge); the facade rejects → warning, children unchanged.
Equal spheres with centers in one plane	not a special case (flat hull of the centers); warning.
Collision with STL meshes	yes/no only, no intersection volume.
Keepout check in the worker	Booleans with filleted bodies block the worker for a few seconds the first time (hole plate ≈ 2.5 s); afterwards cache.
Point contact	detected, but without location (no marking).
STEP: overlapping top-level objects	are written as overlapping individual bodies (notice in the export). User decision pending; proposal: unite overlapping objects into one body, keep touching ones separate.
STL B-rep over 500 faces	Booleans superlinear (curve in section 3); limit automatic, can be forced with `as = "solid"`.
UI settings of the desktop app	settings and theme live in the WebView's `localStorage`, which Tauri stores under the app ID; with `dev.mcad.desktop` they start fresh. Only the list of recently opened files is migrated.
Arch package `mcad`	PKGBUILD, `docker.sh` and `smoke-test.sh` updated, but not built: no `makepkg` and no running Docker daemon in the container.
`compare.py --self-test`	needs reference STLs from OpenSCAD, which are missing here; `--eval` is checked.